Liu, Chapter 6: Sheaves of differentials

Solutions and corrections to the exercises

Conventions and scope. Labels refer to the four exercise lists in Chapter 6 of the supplied 2002 text (printed pp. 218–251). Statements are restated, with subparts in running prose. Write $\Omega_{B/A} = \Omega^1_{B/A}$, and similarly for schemes. A function field means a finitely generated field extension. A regular sequence in a local ring is contained in its maximal ideal. Standard results used below include the conormal and transitivity sequences, the local criterion for flatness, Nakayama's lemma, the Koszul criterion for a regular sequence, and cohomology and base change.

Corrections and limitations. Some requested conclusions are false under the printed hypotheses. We give counterexamples in 2.2, 2.3, 2.7, 3.3 (the global assertion), 3.4, 3.5 and 4.7(b), and prove corrected versions. In 1.1 "module" must mean "algebra", in 1.5 a base subscript is missing, in 2.9 the field extension is reversed, and in 4.11(b) n must be d. Exercise 4.5 needs a larger class of ambient schemes to make its proposed direct-limit construction work. Exercise 2.11 also has a false monogenicity requirement over finite residue fields. Exercise 2.7 needs separability of the specified residue field; we give a counterexample and prove a corrected local version. Exercise 1.3 is proved with B Noetherian; the unrestricted non-Noetherian assertion in (a) is not claimed here. In 4.10 we give the coherent, locally Noetherian base-change statement; arbitrary non-Noetherian base changes are outside the stated scope of these notes. These qualifications are part of the solutions, not claims that the uncorrected statements have been proved.

6.1 Kahler differentials

Exercise 1.1

> **Problem.** Let B be an A-algebra, $\rho : B \otimes_A B \to B$ multiplication, and $I = \operatorname{Ker}\rho$. Give I/I^2 the B-action from the right factor and put $db = b \otimes 1 - 1 \otimes b$ modulo I^2. (a) Prove that d is an A-derivation. (b) Prove its universal property. (c) For a morphism $f : X \to Y$ and its diagonal Δ, let $\mathcal{I} = \operatorname{Ker}(\mathcal{O}_{X\times_Y X} \to \Delta_*\mathcal{O}_X)$. Prove $\Omega_{X/Y} \simeq \Delta^*(\mathcal{I}/\mathcal{I}^2)$.

Solution. (a) Additivity and $da = 0$ for $a \in A$ follow from the tensor relations. The difference $d(bc) - b\,dc - c\,db$ is the product $(b \otimes 1 - 1 \otimes b)(c \otimes 1 - 1 \otimes c)$, hence is zero modulo I^2.

(b) For an A-derivation $D : B \to M$, define $h : B \otimes_A B \to M$ by $h(b \otimes c) = cD(b)$. This is well defined and right B-linear. For $u, v \in B \otimes_A B$, direct expansion of the Leibniz rule gives

$h(uv) = \rho(u)h(v) + \rho(v)h(u)$. Thus $h(I^2) = 0$ and its restriction gives a map $I/I^2 \to M$ sending db to $D(b)$. Every $u = \sum b_i \otimes c_i \in I$ equals $\sum(1 \otimes c_i)(b_i \otimes 1 - 1 \otimes b_i)$, so these elements generate I, proving uniqueness.

(c) On $U = \operatorname{Spec} B \to V = \operatorname{Spec} A$, the diagonal inside $U \times_V U$ has precisely this ideal. The isomorphism in (b) is compatible with restriction and localization and therefore glues. Such product opens cover a neighborhood of the diagonal, which suffices for its pullback. No separatedness hypothesis is needed. □

Exercise 1.2

> **Problem.** Show that $\Omega_{A[[T]]/A}$ need not be finitely generated. Prove that formal differentiation induces an exact sequence $0 \to \bigcap_{n\geq 1} T^n\Omega_{A[[T]]/A} \to \Omega_{A[[T]]/A} \xrightarrow{\varphi} A[[T]]dT$ with $\varphi(dF) = F'dT$.

Solution. Put $R = A[[T]]$ and $M = \Omega_{R/A}$. Formal differentiation is an A-derivation, so defines φ, which is surjective since $\varphi(dT) = dT$. Write $F = P + T^{n+1}G$ with $P \in A[T]$. Then

$$dF - F'dT = T^{n+1}(dG - G'dT) \in T^nM.$$

Since M is generated by the dF, this shows that $M/T^nM \to (R/T^nR)dT$ is an isomorphism: its inverse sends dT to dT. Consequently $\operatorname{Ker}\varphi \subseteq \bigcap_n T^nM$. The reverse inclusion follows from $\bigcap_n T^nR = 0$.

For non-finiteness take $A = \mathbf{Q}$. Localization identifies $M \otimes_R \mathbf{Q}((T))$ with $\Omega_{\mathbf{Q}((T))/\mathbf{Q}}$. The field $\mathbf{Q}((T))$ has infinite transcendence degree over $\mathbf{Q}$: any field of finite transcendence degree over a countable field is countable, whereas $\mathbf{Q}((T))$ is uncountable. In characteristic zero the differentials of a transcendence basis form a basis of the differential module after the remaining algebraic extension. Its dimension is therefore infinite, ruling out finite generation of M. □

Exercise 1.3

> **Problem.** For an ideal $I \subset B$ and its adic completion $\widehat{B}$, let $\rho : \Omega_{B/A} \to \Omega_{\widehat{B}/A}$. (a) Show $\Omega_{\widehat{B}/A} = \rho(\Omega_{B/A}) + I\Omega_{\widehat{B}/A}$ and obtain surjections modulo I^n with kernels K_n satisfying $K_{n+1} \twoheadrightarrow K_n$. (b) If A is Noetherian and B is a localization of a finite type A-algebra, identify $\varprojlim_n \Omega_{\widehat{B}/A}/I^n\Omega_{\widehat{B}/A}$ with $\Omega_{B/A} \otimes_B \widehat{B}$.

Solution. Scope of (a): assume B Noetherian. This guarantees $\widehat{B}/I^j\widehat{B} = B/I^j$. Put $M = \Omega_{B/A}$ and $N = \Omega_{\widehat{B}/A}$. In fact $M/I^nM \to N/I^nN$ is an isomorphism. Here is a useful proof which also controls the kernels. If E is a B/I^n-module, any derivation $D : B \to E$ annihilates I^{n+1} by the product rule. It therefore extends uniquely to $\widehat{B}$ through

$$\widehat{B} \longrightarrow B/I^{n+1},$$

using the derivation on B/I^{n+1} induced by D. Conversely a derivation on $\widehat{B}$ into E annihilates $(I\widehat{B})^{n+1}$, so is determined by its restriction to B. The two differential modules modulo I^n represent the same functor on B/I^n-modules. This proves the isomorphism, and in particular

$K_n = 0$ and the requested transition surjectivity. At $n = 1$ it also proves the displayed equality: coefficients modulo I lift from B.

(b) The hypotheses imply that B is Noetherian and M is finite. Taking inverse limits gives $\widehat{M}$, and the completion theorem for finite modules gives $\widehat{M} = M \otimes_B \widehat{B}$. For a general non-Noetherian B the identification of completion quotients used above requires additional hypotheses; no proof of that unrestricted case is supplied. □

Exercise 1.4

> **Problem.** If $X \to Y$ is a morphism and $Y \to Z$ an open or closed immersion, prove that $\Omega_{X/Z} \to \Omega_{X/Y}$ is an isomorphism.

Solution. For an open immersion work in an open subset of Z identified with Y. For a closed immersion, write the maps locally as $A \to A/J \to C$. An A-derivation on C is precisely an A/J-derivation: the images of elements of A/J are images of elements of A. Thus the two universal differential modules agree, compatibly with restriction. □

Exercise 1.5

> **Problem.** For S-schemes X, Y with product projections p, q, prove $\Omega_{(X \times_S Y)/S} \simeq p^*\Omega_{X/S} \oplus q^*\Omega_{Y/S}$. (The left-hand base subscript is omitted in the print.)

Solution. On affine charts set $R = B \otimes_A C$. A derivation $R \to M$ is uniquely specified by derivations $D_B : B \to M$, $D_C : C \to M$, and is given by $D(b \otimes c) = cD_B(b) + bD_C(c)$. Hence its representing module is $(\Omega_{B/A} \otimes_B R) \oplus (\Omega_{C/A} \otimes_C R)$. These isomorphisms are natural and glue. □

Exercise 1.6

> **Problem.** Let $K \subseteq L$ be function fields separable over k. Show that $\Omega_{K/k} \otimes_K L \to \Omega_{L/k}$ is injective if and only if L/K is separable.

Solution. Write $a = \operatorname{trdeg}_k K$, $b = \operatorname{trdeg}_k L$. By Prop. 6.1.15 the first two modules have dimensions a, b. Transitivity shows that injectivity is equivalent to $\dim_L \Omega_{L/K} = b - a = \operatorname{trdeg}_K L$. By the same proposition this is precisely separability of L/K. □

Exercise 1.7

> **Problem.** If K is perfect of characteristic $p > 0$ and L/K is any field extension, prove $K \subseteq L^p$ and $\Omega_{L/K} \simeq \Omega_{L/L^p}$.

Solution. Every element of K is a pth power in K, hence in L. Every derivation in characteristic p kills pth powers. Thus a derivation on L kills K if and only if it kills L^p: one direction follows from the containment, the other from the product rule. Apply the universal property. □

Exercise 1.8

> **Problem.** Let X be smooth of dimension d over K. (a) For a function field L/K put $r = \dim_L \Omega_{L/K}$; show that L can be generated over K by $r+1$ elements. (b) Deduce that $k(x)$ can be generated by $d+1$ elements for every $x \in X$.

Solution. (a) Choose $u_1, \dots, u_r$ with du_i a basis and set $E = K(u_1, \dots, u_r)$. Transitivity gives $\Omega_{L/E} = 0$. A finitely generated field extension with zero differential module is finite separable (Prop. 6.1.15 followed by Lemma 6.1.13(c)). The primitive element theorem gives $L = E(v)$, proving the assertion, without requiring that the u_i be algebraically independent.

(b) The conormal sequence for $\mathcal{O}_{X,x} \to k(x)$ surjects $\Omega_{X/K} \otimes k(x)$ onto $\Omega_{k(x)/K}$. The source has dimension at most d, so (a) applies with $r \le d$. □

Exercise 1.9

> **Problem.** Let $S = \operatorname{Spec} \mathbf{Z}_p$ and $X = \operatorname{Spec} \mathbf{Z}_p[u,v]/(uv^2+p)$. Prove that X is regular and integral and that $(\Omega_{X/S})_{\mathrm{tors}}$ is nonzero coherent with support $V(v)$.

Solution. Put $R = \mathbf{Z}_p[u,v]/(uv^2+p)$. Gauss's lemma applied over $\mathbf{Z}_p[v]$ shows that the primitive linear polynomial v^2u+p is irreducible, hence prime in this UFD; thus R is a domain. On $D(v)$, $R_v = \mathbf{Z}_p[v, v^{-1}]$ is regular. A prime containing v contains p; in the regular ambient local ring of $\mathbf{Z}_p[u,v]$ at that prime, p is part of a regular system of parameters and $uv^2 \in \mathfrak{m}^2$. Thus $p + uv^2 \notin \mathfrak{m}^2$, and its quotient is regular. This covers X.

Now

$$\Omega_{R/\mathbf{Z}_p} = (R\,du \oplus R\,dv)/R\,v(v\,du + 2u\,dv).$$

Set $w = v\,du + 2u\,dv$. The saturation of the relation depends on whether $p = 2$, so we separate the cases.

If $p \neq 2$, the vector w is primitive at every height-one prime: the unique height-one prime containing v is (v), and $2u$ is a unit there. Since R is normal, its saturated span in R^2 is Rw. Consequently the torsion is $Rw/Rvw \simeq R/(v)$. If $p = 2$, the relation becomes $v^2(du - u^2 v\,dv) = 0$, since $2 = -uv^2$. The vector $du - u^2 v\,dv$ is part of a basis of R^2, so the torsion is $R/(v^2)$ times this vector. In both cases it is nonzero, coherent, and has exactly the asserted support. □

6.2 Differential study of smooth morphisms

Exercise 2.1

> **Problem.** If $X \to S$ is smooth and $x \in X_s$ is $k(s)$-rational, prove $\widehat{\mathcal{O}}_{X,x} \simeq \widehat{\mathcal{O}}_{S,s}[[T_1, \ldots, T_n]]$ as complete base algebras, with $n = \dim \mathcal{O}_{X_s,x}$.

Solution. Choose relative functions vanishing at x whose differentials form a basis. They define an *etale* map from a neighborhood of x to $\mathbf{A}^n_S$, taking x to the origin over s (Cor. 6.2.11). An etale local map with identical residue fields induces an isomorphism on completed local rings: successive lifts across powers of the maximal ideal are unique by formal etaleness. The completed local ring at that origin is $\widehat{\mathcal{O}}_{S,s}[[T_1, \ldots, T_n]]$. This also respects the base map. □

Exercise 2.2

> **Problem.** For an algebraic variety X/k, the print asks to prove: X is smooth if and only if $\Omega_{X/k}$ is locally free and every generic residue field $k(\xi)/k$ is separable.

Solution. A generically reduced hypothesis is needed. Liu's varieties are finite type schemes, not necessarily reduced. In characteristic p, $X = \operatorname{Spec} k[t]/(t^p)$ has free differential module $\mathcal{O}_X dt$, and its only residue field is k, but is not smooth. Thus the printed statement is false.

Here is the corrected assertion for generically reduced X (in particular for reduced varieties). Smoothness implies both conditions. Conversely, near a point where Ω is free of rank r, each irreducible component has dimension r: at its generic point the local ring is a field by generic reducedness, and separability identifies the differential rank with its transcendence degree. Thus $\dim_x X = r$ throughout this neighborhood. Proposition 6.2.2, the criterion that $\Omega_{X/k,x}$ be free of rank $\dim_x X$, now gives smoothness. □

Exercise 2.3

> **Problem.** The print asserts that if $X \to S$ is smooth, then $s \mapsto \dim X_s$ is locally constant.

Solution. False. Let $S = \mathbf{A}^1_k$, $U = D(t)$, and $X = S \amalg \mathbf{A}^1_U$ with its natural smooth surjective morphism to S. The fibre has dimension 1 on U and dimension 0 at 0.

The valid general statement is that $x \mapsto \dim_x X_{f(x)}$ is locally constant on X: it equals the rank of the locally free sheaf $\Omega_{X/S}$. If the morphism is proper and smooth, its nonempty fibre dimensions are locally constant on its image. Indeed, near any base point only finitely many relative-rank strata occur; each is open and closed in X, and its image is open by smoothness and closed by properness. The maximum of the ranks present is therefore locally constant. □

Exercise 2.4

Problem. For a smooth connected variety X/k in characteristic $p > 0$ and relative Frobenius $F : X \to X^{(p)}$, compute $\Omega_{X/X^{(p)}}$: (a) for affine space and (b) in general.

Solution. Locally the relative Frobenius homomorphism is $B \otimes_{k,F_k} k \to B$, $b \otimes c \mapsto b^p c$. Every k-derivation kills its image, and every derivation relative to that image kills k. Hence, canonically,

$$\Omega_{X/X^{(p)}} = \Omega_{X/k}.$$

For affine space the basis is $dT_1, \ldots, dT_n$. In general smoothness makes this sheaf locally free; on connected X its rank is constant and equals $\dim X$. This argument does not require that k be perfect. □

Exercise 2.5

Problem. For $X \to S$ of finite type and $x \in X_s$, (a) Prove

$$\dim_{k(x)} T_{X_s,x} \leq \dim_{k(x)}(\Omega_{X_s/k(s),x} \otimes k(x)).$$

(b) Give the exact sequence

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \to \Omega_{X_s/k(s),x} \otimes k(x) \to \Omega_{k(x)/k(s)} \to 0.$$

(c) If x is closed in X_s with separable residue field, identify the dual of $\Omega_{X/S} \otimes k(x)$ with $T_{X_s,x}$.

Solution. (a) By Lemma 6.2.4, near x there is a closed immersion $X \to Z$ with Z/S smooth of relative dimension $d = \dim_{k(x)} \Omega_{X/S} \otimes k(x)$. Closed immersions inject Zariski tangent spaces. The embedding dimension of the regular local ring $\mathcal{O}_{Z_s,x}$ equals its dimension, which is at most d. This proves the inequality, including at nonclosed points.

(b) Put $R = \mathcal{O}_{X_s,x}$ and $\mathfrak{n} = \mathfrak{m}_R$. The conormal sequence for $R \to k(x)$ is

$$\mathfrak{n}/\mathfrak{n}^2 \longrightarrow \Omega_{R/k(s)} \otimes_R k(x) \longrightarrow \Omega_{k(x)/k(s)} \longrightarrow 0.$$

The map $\mathfrak{m}_x/\mathfrak{m}_x^2 \to \mathfrak{n}/\mathfrak{n}^2$ is surjective, giving exactly the requested sequence.

(c) Now $k(x)/k(s)$ is finite separable. In addition the first map from $\mathfrak{n}/\mathfrak{n}^2$ is injective: a finite separable residue extension has a coefficient-field lift into $R/\mathfrak{n}^2$ by formal etaleness, and the resulting split square-zero algebra shows that its relative differentials tensored with $k(x)$ are $\mathfrak{n}/\mathfrak{n}^2$. Thus $\Omega_{X/S} \otimes k(x) \simeq \mathfrak{n}/\mathfrak{n}^2$. Dualizing gives the tangent space. □

Exercise 2.6

> **Problem.** Let $X \to Y$ be of finite type between locally Noetherian regular schemes, and let $x \in X_y$ be closed with $k(x)/k(y)$ separable. (a) Locally embed X into a smooth Y-scheme of relative dimension $\dim \mathcal{O}_{X,x}$. (b) Locally realize X as a closed subscheme of codimension $\dim \mathcal{O}_{Y,y}$ in a smooth Y-scheme.

Solution. Set $a = \dim \mathcal{O}_{Y,y}$ and $b = \dim \mathcal{O}_{X,x}$. By 2.5(c), the dimension e of $\Omega_{X/Y} \otimes k(x)$ is the embedding dimension of the fibre local ring and is at most b, since that fibre ring is a quotient of $\mathcal{O}_{X,x}$. Lemma 6.2.4 embeds a neighborhood in a smooth Y-scheme of relative dimension e. Take its product with $\mathbf{A}_Y^{b-e}$ and embed X using zero in the new coordinates. This proves (a).

The ambient local ring at x is regular of dimension $a+b$, since x is closed in its fibre and has finite residue extension. Its quotient $\mathcal{O}_{X,x}$ is regular of dimension b. The kernel is therefore generated by a members of a regular system of parameters: lift a basis of the kernel on cotangent spaces and complete it to a parameter basis. The quotient by those a parameters is a regular local domain of dimension b, so its further prime quotient of dimension b has zero kernel. Spreading these generators to a neighborhood proves (b). □

Exercise 2.7

> **Problem.** Let $X \to S$ be finite type and smooth at a point x closed in X_s. (a) If $\dim X_s > 0$, find a closed subscheme Z through x, smooth at x, with smaller special-fibre dimension. (b) Find an etale $T \to S$ such that $X_T \to T$ has a section whose composite to X contains x.

Solution. Both parts need a separability hypothesis on $k(x)/k(s)$. Let $k = \mathbf{F}_p(a)$, $S = \operatorname{Spec} k$, $X = \mathbf{A}_k^1$, and take the closed point defined by $T^p - a$. Its residue field is purely inseparable of degree p. A zero-dimensional closed subscheme through this point cannot be smooth over k: a smooth zero-dimensional k-scheme has finite separable residue fields. Thus (a) is false. Similarly an etale $T \to \operatorname{Spec} k$ cannot have a point mapping to x, since that would embed $k(x)$ in a finite separable extension of k. This disproves (b).

For a corrected version assume $k(x)/k(s)$ finite separable. Locally at x, if the relative dimension n is positive, 2.5(c) gives a function vanishing at x with nonzero relative differential. Its zero locus on a sufficiently small affine neighborhood is smooth at x of relative dimension $n-1$, by the Jacobian criterion. This proves the useful local form of (a). No bound on the whole fibre of a global closure is asserted here.

For (b) under this extra assumption choose etale coordinates $h : U \to \mathbf{A}_S^n$ near x. Lift the finite separable extension $k(x)/k(s)$ to an etale neighborhood $S' \to S$, and lift the coordinates of $h(x)$ to functions on S' near its distinguished point. These define a section $a : S' \to \mathbf{A}_S^n$ whose distinguished fibre value agrees with $h(x)$ over $k(x)$. The fibre product $T = U \times_{\mathbf{A}_S^n} S'$ is etale over S and has a point mapping to x. Its map $T \to U \to X$ gives the required section of $X_T \to T$. □

Exercise 2.8

Problem. Let S be Noetherian local with closed point s. If $X \to S$ is proper and flat and $X_s/k(s)$ is smooth, prove that X/S is smooth.

Solution. The nonsmooth locus is closed, since a proper morphism is finite type and smoothness is open in this setting. Its image is closed by properness. It misses s, since flatness and geometric regularity of the special fibre give smoothness there. Every nonempty closed subset of a local scheme contains its closed point; hence the image is empty. □

Exercise 2.9

Problem. Let $f : X \to Y$ be dominant of finite type between locally Noetherian integral schemes. Call it generically separable (respectively generically etale) when $K(X)/K(Y)$ is separable (respectively finite separable); the print reverses this extension. (a) Find a dense open smooth (respectively etale) locus. (b) If f is proper and its generic fibre is smooth, find a nonempty open $V \subset Y$ over which f is smooth (respectively etale under the corresponding generic assumption).

Solution. (a) At the generic point of X, the morphism is smooth: choose a separating transcendence basis of $K(X)/K(Y)$ followed by a primitive element for the finite separable part. Clearing denominators gives a presentation etale over an affine space on a nonempty open of X (also shrink the base to clear coefficients). Smoothness is open, so this yields the assertion. If the field extension is finite separable the affine space has dimension zero and the morphism is etale there.

(b) Let C be the closed nonsmooth locus. The whole generic fibre is smooth by assumption, so $f(C)$ is a proper closed subset; take its complement. In the generically etale case the smooth generic fibre is integral and zero-dimensional, hence finite separable and everywhere etale. Use instead the closed non-etale locus. Properness of f is what makes its image closed. □

Exercise 2.10

Problem. For an S-morphism $f : X \to Y$ between smooth S-schemes, prove exactness of $0 \to f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$ (a) if f is smooth, and (b) if X, Y are integral and f is dominant and generically separable.

Solution. (a) Smoothness gives, locally, an etale factorization through affine space over Y. For a polynomial algebra the sequence splits with complementary basis the relative coordinate differentials. Etale pullback preserves this description, proving left injectivity as well as right exactness.

(b) At the generic point, choose an affine base neighborhood $\operatorname{Spec} R \subset S$. Its image in $K(Y)$ is a domain with fraction field k. Since X, Y are smooth over S, localization and base change identify their differential spaces with those of the separably generated fields $K(X), K(Y)$ over k. Exercise 1.6 gives generic injectivity. The kernel is a subsheaf of the locally free sheaf $f^*\Omega_{Y/S}$ on integral X; a section of a locally free sheaf which vanishes generically is zero. Right exactness is transitivity. □

Exercise 2.11

> **Problem.** For a finite map of Noetherian rings $A \to B$ with A complete local and residue field k, let k' be the largest separable k-subalgebra of $B \otimes_A k$. Factor the map through a simple finite etale A-algebra C with $C \otimes_A k = k'$.

Solution. We use two standard properties of finite etale algebras over a complete local ring: reduction modulo the maximal ideal is an equivalence with finite etale k-algebras, and morphisms from such algebras lift uniquely across nilpotent thickenings. Let C lift k'. The inclusion $k' \to B/\mathfrak{m}_A B$ lifts uniquely through $B/\mathfrak{m}_A^n B$. Taking limits gives $C \to B$, since the finite A-module B is complete. No injectivity of this map is asserted or required: the original AoB need not be injective. The algebra B is finite over C because any finite set of A-module generators also generates over C.

For the additional monogenicity requirement, first suppose k is infinite. Then choose primitive elements in its field factors with pairwise coprime minimal polynomials, using translations. If k is finite, the maximal separable subalgebra can have arbitrarily many identical residue-field factors. Thus monogenicity is *not* automatic in this case: for $k = \mathbf{F}_2$, $B = k^3$, and $A = k$, the largest separable subalgebra is k^3, which no single element generates (a single element has only the two possible coordinate values). The finite etale factorization always exists; the additional printed requirement "C simple" is false without an extra monogenicity hypothesis on k', for example k infinite. In that case lift a primitive element; Nakayama shows its lift generates C over A. □

6.3 Local complete intersection

Exercise 3.1

> **Problem.** For a regular sequence $a_1, \ldots, a_n$ in the maximal ideal of a Noetherian local ring, (a) reduce invariance under permutations to two elements, and (b) for $J = \mathrm{Ann}(a_2)$ show $J = a_1 J$, deduce $J = 0$, and prove (a_2, a_1) regular.

Solution. (a) Adjacent transpositions generate the symmetric group. To interchange positions $i, i+1$, work modulo the preceding elements; the quotient by the two elements, and hence the regularity of the remaining tail, is unchanged.

(b) If $a_2 x = 0$, regularity of a_2 modulo a_1 gives $x = a_1 y$. Since a_1 is a nonzerodivisor, $a_2 y = 0$. Thus $J = a_1 J$. This finite ideal is zero by Nakayama. Finally if $a_1 x = a_2 y$, reduction modulo a_1 gives $y = a_1 z$, and cancellation gives $x = a_2 z$. Thus a_1 is regular modulo a_2. □

Exercise 3.2

Problem. Keep the preceding hypotheses. Put $I = (a_1, \dots, a_n)$ and choose minimal generators $b_1, \dots, b_m$. (a) After permuting the a_i, show

$$b_m = ua_n + \sum_{i<n} c_i a_i$$

with u a unit and the resulting sequence regular. (b) Prove $m = n$ and regularity of the b_i. (c) If (J, T) in $A[T]_{(\mathfrak{m},T)}$ is generated by a regular sequence, prove the same for J in A, also for several variables. (d) Prove descent of this property along a faithfully flat Noetherian local extension $A \to C$.

Solution. By Lemma 6.3.6, I/I^2 is free of rank n over A/I, so $\dim_{A/\mathfrak{m}} I/\mathfrak{m}I = n$. Thus $m = n$, and a minimal generator b_m has a unit coefficient in an expression in the a_i. Permute that coefficient to the last place using 3.1. Modulo the first $n-1$ elements, b_m is a unit multiple of a_n, proving (a). More generally the two minimal systems are related by an invertible matrix over A, since their reductions are bases of $I/\mathfrak{m}I$. Their Koszul complexes are isomorphic under this change of basis. Positive Koszul homology vanishes for the a_i, hence for the b_i, which proves (b) by the local Koszul criterion.

(c) A minimal system for J together with T is minimal for (J, T): reduce its cotangent space first modulo T, and observe that the coefficient of T cannot be supplied by $J \subset \mathfrak{m}$. By (b) it is regular, in the order starting with T. Modulo T the remaining elements form a regular sequence in A. Iterate for several variables.

(d) A minimal system $j_1, \dots, j_r$ for J remains minimal after a flat local extension: $JC/\mathfrak{m}_C JC \simeq (J/\mathfrak{m}_A J) \otimes_{k(A)} k(C)$. By (b) it is regular in C. Flat base change of the Koszul complex and faithful flatness imply that its positive homology over A vanishes. Apply the Koszul criterion again. The zero and unit ideals can be handled separately, with the empty sequence for zero. □

Exercise 3.3

Problem. For immersions $X \xrightarrow{f} Y \xrightarrow{g} Z$, the print asks: if f and gf are regular, show g regular; give an example where g and gf are regular but f is not.

Solution. The first assertion is local along $f(X)$. In a local ambient ring write $Y = V(I)$, $X = V(J)$, $I \subseteq J$. Assume J and J/I are regular-sequence ideals. Lift a minimal regular system $b_1, \dots, b_r$ of J/I. Its classes are independent in $J/\mathfrak{m}J$ and can be extended using elements $a_1, \dots, a_c \in I$ to a minimal generating system for J. By 3.2 this system is regular. Since the b_i are regular modulo I, their Koszul syzygies of their images generate all relations. Applying this to expressions for elements of I in (a, b) shows

$$I = (a_1, \dots, a_c) + I(b_1, \dots, b_r).$$

Indeed, subtract the lifted Koszul relations to make every coefficient of b_i lie in I. Nakayama gives $I = (a_1, \dots, a_c)$, and this is a regular sequence as a subsequence of a permutable regular sequence. This proves regularity at points of $f(X)$, hence on a neighborhood of that set.

There is no assertion away from $f(X)$. For a global counterexample take $Z = \mathbf{A}^1_k \amalg \mathbf{A}^2_k$, $Y = \{0\} \amalg V(x^2, xy, y^2)$, and $X = \{0\}$ in the first component. Both f and gf are regular, while g is not regular in the second component, whose defining ideal has height 2 and three minimal generators.

For the requested converse counterexample take $Z = \operatorname{Spec} k[t]$, $Y = V(t^2)$, $X = V(t)$. Both immersions into Z are regular, but $X \to Y$ is defined by the zero divisor t in $k[t]/(t^2)$. □

Exercise 3.4

> **Problem.** Let $(A, \mathfrak{m})$ be Noetherian local and equidimensional. The print asks: (a) a regular sequence of length r has equidimensional quotient of dimension $\dim A - r$; (b) if the next element is a zero divisor, any extension to m elements has quotient dimension at least $\dim A - m + 1$; (c) m elements cutting the dimension down by m form a regular sequence.

Solution. Equidimensionality alone is insufficient for (b) and (c). Take

$$A = k[x, y]_{(x,y)}/(x^2, xy).$$

It has one minimal prime (x) and dimension 1, hence is equidimensional. But y is a zero divisor, since $xy = 0$ with $x \neq 0$, whereas $A/(y) = k[x]_{(x)}/(x^2)$ has dimension 0. This contradicts (b) with $r = 0, m = 1$ and also (c).

A valid version of all three parts assumes A Cohen–Macaulay. A quotient by a regular sequence is Cohen–Macaulay of dimension $\dim A - r$, and a Cohen–Macaulay local ring is equidimensional, proving (a) in this version. Its associated primes are minimal and their quotients have the full dimension. If a_{r+1} is a zero divisor it lies in one such prime $\mathfrak{p}$. The remaining $m - r - 1$ elements decrease $\dim A/\mathfrak{p}$ by at most that number, by the height theorem. The resulting quotient is a quotient of $A/(a_1, \ldots, a_m)$ and has dimension at least $\dim A - m + 1$. This proves (b). For (c), a first failure of regularity would contradict (b). Equivalently, extend the b_i to a system of parameters; in a Cohen–Macaulay local ring every system of parameters is regular. The unrestricted assertion (a) is not used in these notes; its equidimensional conclusion also requires care without catenarity hypotheses. □

Exercise 3.5

> **Problem.** Let $f : X \to Y$ be finite surjective and l.c.i. The print asks: (a) $\dim \mathcal{O}_{X,x} = \dim \mathcal{O}_{Y,f(x)}$; (b) locally present X in $\mathbf{A}^n_V$ by n equations; (c) show the fibre equations regular and deduce flatness.

Solution. These conclusions are false as stated. Let $Y = \operatorname{Spec} k[t]$ and $X = Y \amalg V(t)$, mapping by the identity and the closed immersion. This is finite, surjective and l.c.i. On the second component at its unique point the local dimensions are 0 and 1. It is not flat, since that component has coordinate module $k[t]/(t)$.

The correct local criterion is: a finite l.c.i. of *virtual relative dimension zero* is flat. To see this, factor near a point as a regular immersion in a smooth scheme of relative dimension n. Virtual dimension zero means that its ideal is generated by a regular sequence of length n;

with affine-space presentations the same equality gives (b). At a point of the finite fibre, the ambient fibre ring is a regular local ring of dimension n, and the quotient by the n equations has dimension zero. Thus those equations form a system of parameters in a Cohen–Macaulay ring, hence a regular sequence. The local flatness criterion applied to the smooth ambient algebra and its fibrewise regular equations proves flatness. It then yields the dimension formula in (a). Virtual dimension zero, missing from the printed hypotheses, is essential; the counterexample has virtual dimension -1 on its extra component. □

Exercise 3.6

> **Problem.** If a closed immersion $X \to Y$ has ideal sheaf $\mathcal{J}$ whose stalk at x is generated by a regular sequence, show that this is true on an affine neighborhood of x in Y.

Solution. Lift the finite regular system to sections $a_1, \dots, a_r$ on an affine neighborhood. Coherence of $\mathcal{J}$ allows us to shrink so that they generate it: the cokernel of their map to $\mathcal{J}$ has closed support missing x. For each i the annihilator of a_i in $\mathcal{O}_Y/(a_1, \dots, a_{i-1})$ is coherent and has zero stalk at x. Delete the union of these finitely many supports and choose a smaller affine neighborhood. On its ring of sections multiplication by every a_i is injective on the preceding quotient, as can be checked on stalks. Thus the sections form the required regular sequence. □

Exercise 3.7

> **Problem.** Let $X \to Y$ be flat finite type, $x \in X_y$ rational over $k(y)$, with fibre l.c.i. at x. (a) Present $\widehat{\mathcal{O}}_{X_y,x}$ as $k(y)[[T_1, \dots, T_n]]/(f_1, \dots, f_r)$ with a regular sequence, allowing $n = \dim_{k(x)} \Omega_{X_y/k(y)} \otimes k(x) + 1$. (b) Lift any such presentation to
>
> $$\widehat{\mathcal{O}}_{X,x} = \widehat{\mathcal{O}}_{Y,y}[[T_1, \dots, T_n]]/(F_1, \dots, F_r),$$
>
> with F_i reducing to f_i.

Solution. (a) A local l.c.i. presentation in a smooth ambient scheme completes to a power-series presentation, by 2.1. Faithfully flat completion preserves its regular sequence. If e is the embedding dimension of the fibre ring, eliminate any equation having a nonzero linear term by a formal change of coordinates and formal substitution. Repeating leaves a presentation in e variables, with regular kernel. Adjoining one new variable and its equation gives the printed $n = e + 1$.

(b) Put $A = \widehat{\mathcal{O}}_{Y,y}$, $C = \widehat{\mathcal{O}}_{X,x}$, and $P = A[[T_1, \dots, T_n]]$. Lift the images of T_i in the special-fibre completion to the maximal ideal of C. The resulting continuous map $P \to C$ is surjective by successive approximation: the base maximal ideal and those images generate the maximal ideal of C, and the residue fields agree. The ring C is flat over A, by the completion theorem for a flat local map of Noetherian rings. If J is the kernel, flatness gives

$$J/\mathfrak{m}_A J = \operatorname{Ker}(P/\mathfrak{m}_A P \to C/\mathfrak{m}_A C) = (f_1, \dots, f_r).$$

Choose lifts $F_i \in J$ of the prescribed f_i. Nakayama applied to the finite P-module J gives $J = (F_1, \dots, F_r)$. The fibrewise regular-sequence lifting lemma (the local flatness criterion, applied successively) shows the F_i regular in the A-flat ring P. This proves the asserted presentation with exactly the prescribed reductions. □

Exercise 3.8

Problem. Let $f : X \to Y$ be a dominant l.c.i. of integral Noetherian S-schemes. Suppose $K(X)/K(Y)$ is separable. (a) in a regular immersion $X \to Z = \mathbf{A}^n_Y$, prove injectivity of $\delta : \mathcal{C}_{X/Z} \to \Omega_{Z/Y}|_X$; (b) prove the transitivity sequence left exact:

$$0 \to f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0.$$

Solution. (a) The source is locally free by regularity of the immersion. At the generic point of X, the map f is smooth by 2.9(a). Its conormal sequence for a smooth ambient presentation is left exact there, so δ is generically injective. Its kernel, being a subsheaf of a locally free sheaf on an integral scheme, is zero.

(b) Locally write $B = A[T_1, \ldots, T_n]/I$ over a base ring R. The absolute conormal map has components in

$$(\Omega_{A/R} \otimes_A B) \oplus B^n.$$

The second component is precisely the injective relative conormal map $I/I^2 \to B^n$ from (a). Therefore the image of I/I^2 intersects the first summand trivially. Taking the quotient gives injectivity of $\Omega_{A/R} \otimes_A B \to \Omega_{B/R}$. Transitivity gives the remaining exactness, and these local sequences glue. □

6.4 Duality theory

Exercise 4.1

Problem. For $f : X \to Y$ finite type over locally Noetherian Y, suppose $\Omega^r_{X/Y} = 0$ with $r \geq 1$. (a) Show a finite module M over a local ring with $\bigwedge^r M = 0$ has at most $r - 1$ generators. (b) If $df_1, \ldots, df_n$ generate $\Omega_{B/A}$, show $\Omega_{B/A[T_1,\ldots,T_n]} = 0$ for $T_i \mapsto f_i$. (c) Locally immerse X into $\mathbf{A}^r_Y$.

Solution. (a) Exterior powers commute with base change. Hence $\bigwedge^r(M/\mathfrak{m}M) = 0$, so this residue vector space has dimension less than r. Lift a basis and use Nakayama.

(b) Transitivity identifies the relative differential module with the cokernel of the map whose image is generated by the df_i; that cokernel is zero.

(c) At a chosen point, (a) lets us choose $n \leq r - 1$ functions whose differentials generate, after shrinking. The resulting map $X \to \mathbf{A}^n_Y$ is unramified by (b). The local structure theorem for unramified morphisms factors it locally as a closed immersion into a standard etale scheme over $\mathbf{A}^n_Y$. A standard etale scheme is an open subscheme of a hypersurface in $\mathbf{A}^{n+1}_Y$. Thus X immerses locally into $\mathbf{A}^{n+1}_Y$, and composing with zero coordinates gives the required immersion into $\mathbf{A}^r_Y$. □

Exercise 4.2

Problem. For $F = x^n + y^n + z^n$ with $n \geq 1$ invertible in k, determine $H^0(X, \Omega_{X/k})$ for $X = V_+(F) \subset \mathbf{P}^2_k$.

Solution. The partial derivatives have no common projective zero, so X is smooth. Adjunction gives $\Omega_{X/k} = \omega_{X/k} \simeq \mathcal{O}_X(n-3)$. The restriction sequence

$$0 \to \mathcal{O}_{\mathbf{P}^2}(-3) \xrightarrow{F} \mathcal{O}_{\mathbf{P}^2}(n-3) \to \mathcal{O}_X(n-3) \to 0$$

and $H^1(\mathbf{P}^2, \mathcal{O}(-3)) = 0$ identify the requested space with $k[x,y,z]_{n-3}$, interpreted as zero for negative degree. Its dimension is $(n-1)(n-2)/2$ for $n \geq 3$ and zero for $n = 1, 2$.

Explicitly for $n \geq 3$, a basis is given on $z = 1$ by

$$x^a y^b \frac{dx}{n y^{n-1}}, \qquad a, b \geq 0, \quad a + b \leq n - 3.$$

Where the denominator vanishes use $-x^a y^b dy/(n x^{n-1})$; these agree by $dF = 0$, and adjunction ensures regularity also along infinity. □

Exercise 4.3

Problem. For locally free sheaves $\mathcal{F}, \mathcal{G}$ of ranks m, n, prove $\det(\mathcal{F} \otimes \mathcal{G}) \simeq (\det \mathcal{F})^{\otimes n} \otimes (\det \mathcal{G})^{\otimes m}$.

Solution. Choose local bases e_i, f_j and order the tensor basis lexicographically. Send its top wedge to $(e_1 \wedge \cdots \wedge e_m)^{\otimes n} \otimes (f_1 \wedge \cdots \wedge f_n)^{\otimes m}$. Under basis changes P, Q, the tensor basis changes by $P \otimes Q$, whose determinant is $(\det P)^n (\det Q)^m$. This identity holds over all commutative rings, as a polynomial identity over $\mathbf{Z}$. Thus the local maps glue. The zero-rank cases follow from the convention $\det 0 = \mathcal{O}_X$. □

Exercise 4.4

Problem. Let X/k be normal of pure dimension d, quasi-projective and l.c.i., with k perfect. (a) For $A = \mathcal{O}_X(U)$ on an affine open and a nonzero ideal I, show every A-linear map $I \to A$ is multiplication by $\theta \in \operatorname{Frac} A$; if $\operatorname{ht} I \geq 2$, prove $A^\vee \simeq I^\vee$. (b) For $d = \dim X$, prove $(\Omega^d_{X/k})^{\vee\vee} \simeq \omega_{X/k}$ via the fundamental-class map.

Solution. Work componentwise if necessary; the assertion with a fraction field is for an integral component. The pure-dimension convention is needed for the use of c in (b); for components of different dimensions use their respective dimensions. For $0 \neq a \in I$ and $b \in I$, linearity gives $a\phi(b) = \phi(ab) = b\phi(a)$, so $\phi(b) = \theta b$ with $\theta = \phi(a)/a$. At a prime not containing I, $I_\mathfrak{p} = A_\mathfrak{p}$ implies $\theta \in A_\mathfrak{p}$. If $\operatorname{ht} I \geq 2$ this holds at every height-one prime, and normality gives $A = \bigcap_{\operatorname{ht}\mathfrak{p}=1} A_\mathfrak{p}$ inside its fraction field. Thus $\theta \in A$ and restriction of linear forms is an isomorphism, proving (a).

(b) Normality gives regularity in codimension one, and over a perfect field these points are smooth. The map $c : \Omega^d_{X/k} \to \omega_{X/k}$ is an isomorphism on that smooth locus. Locally trivialize ω and write $\operatorname{Im} c = I \subset A$. The quotient A/I is supported in codimension at least two. The kernel of c is torsion, so any map from Ω^d to A kills it. Hence $(\Omega^d)^\vee = I^\vee = A$ by (a). Dualizing again identifies the induced bidual map with $A \to A$, an isomorphism. These identifications glue. □

Exercise 4.5

> **Problem.** For factorizations $X \xrightarrow{i_k} Z_k \xrightarrow{g_k} Y$ with regular i_k and smooth g_k, put $\omega_{i_k,g_k} = \det \mathcal{C}^\vee_{X/Z_k} \otimes i_k^* \det \Omega_{Z_k/Y}$. Prove: (a) comparisons reduce to smooth maps $Z_3 \to Z_2 \to Z_1$ compatible with X; (b) complete local differential bases $d\alpha$ by $d\beta$ with β in the ideal of X; (c) the comparison sends $a^* \otimes d\alpha$ to $(a^* \wedge \beta^*) \otimes (d\alpha \wedge d\beta)$; (d) deduce the cocycle rule; (e) construct $\omega_{X/Y}$ as a direct limit over projective-space factorizations and prove restriction compatibility.

Solution. (a) Use the product $Z_1 \times_Y Z_2$ and the map (i_1, i_2) from X. Its projections to the two ambients are smooth, and its immersion is regular by Cor. 6.3.22. A triple product gives a common comparison for three choices. It is enough to compute on towers of these smooth comparisons.

(b) The conormal sequence for the two embeddings gives

$$0 \to \mathcal{C}_{X/Z_1} \to \mathcal{C}_{X/Z_2} \to i_2^* \Omega_{Z_2/Z_1} \to 0.$$

Locally choose β_j in the second ideal lifting a basis of the right-hand bundle. Their differentials form a relative basis along X, hence near the chosen point by Nakayama. Smooth transitivity shows that $d\alpha, d\beta$ is a basis of $\Omega_{Z_2/Y}$ there.

(c) The same exact sequence says that the classes of a_i, β_j are a conormal basis. In the determinant comparison the extra conormal determinant and the extra differential determinant cancel. With the ordered-basis convention in the question the formula is therefore exactly

$$a^* \otimes d\alpha \longmapsto (a^* \wedge \beta^*) \otimes (d\alpha \wedge d\beta).$$

A change in the complementary basis multiplies one determinant by a unit and the dual determinant by its inverse, so the formula is independent of the lifts.

(d) For a further smooth comparison choose γ in the third ideal lifting its relative differential basis, as in (b). Applying (c) twice yields

$$(a^* \wedge \beta^* \wedge \gamma^*) \otimes (d\alpha \wedge d\beta \wedge d\gamma).$$

This equals the direct comparison by associativity of exterior multiplication. Thus $\phi_{13} = \phi_{23}\phi_{12}$; the product reductions in (a) give this for arbitrary triples.

(e) **The proposed index category needs a correction.** Products of projective spaces are not projective spaces, and their smooth projections do not furnish the required transition maps from projective-space ambients. Indeed over a field a morphism $\mathbf{P}^N \to \mathbf{P}^m$ with $N > m > 0$ cannot be a nonconstant smooth morphism: a basepoint-free positive-degree system of $m + 1$

forms on $\mathbf{P}^N$ would have a common zero. Thus one cannot simply call the printed collection directed.

Use instead the groupoid of smooth ambient factorizations, locally around X, with the canonical pairwise comparisons constructed using products. Parts (a)–(d) show these comparisons are independent and satisfy the cocycle law. They identify all the invertible sheaves canonically; their glued value defines $\omega_{X/Y}$. Equivalently take the colimit of this system of canonical isomorphisms after choosing a set of representatives. Restricting the factorizations to an open $U \subset X$ restricts the same formulas, so $\omega_{X/Y}|_U \simeq \omega_{U/Y}$. An open subscheme of a quasi-projective scheme is quasi-projective, using the locally finite type notion here. This supplies the intended construction without the unsupported directedness assertion. □

Exercise 4.6

> **Problem.** Extend the construction of $\omega_{X/Y}$ to any l.c.i. over locally Noetherian Y: (a) glue from quasi-projective opens; (b) prove restriction to arbitrary opens; (c) identify it with $\det \mathcal{C}_{X/Z}^{\vee} \otimes i^* \det \Omega_{Z/Y}$ for any regular/smooth factorization; (d) extend the flat base-change and composition properties of Theorem 6.4.9.

Solution. (a) Cover X by affine opens mapping into affine opens of Y. Each has a smooth ambient factorization. The comparison isomorphisms of 4.5 agree on overlaps, and their cocycle law gives descent data for invertible sheaves. Gluing gives $\omega_{X/Y}$ with its specified local identifications. Uniqueness means uniqueness *compatible with these identifications*; the existence of unspecified isomorphisms alone would not be a uniqueness statement.

(b) Restrict that same cover and descent data. Their glued value is precisely $\omega_{U/Y}$, proving the canonical restriction isomorphism.

(c) Compare the given factorization locally with those of the cover. The comparison formulas identify the local determinant expression with the glued bundle and agree on intersections by 4.5(d).

(d) Flat base change preserves the regular sequences defining the immersion, commutes with conormal bundles and relative differentials, and commutes with determinants. This proves the flat base-change isomorphism locally, hence globally. If f itself is flat, the same argument applies to any base change with locally Noetherian target: the fibrewise regular-sequence criterion preserves the immersion and identifies its pulled-back conormal bundle with the new conormal bundle. This is the other case of Theorem 6.4.9(b). For a composition of l.c.i. morphisms, choose compatible local smooth ambient factorizations. The conormal and smooth transitivity exact sequences, followed by determinant cancellation, give

$$\omega_{X/Z} \simeq \omega_{X/Y} \otimes f^* \omega_{Y/Z}.$$

The associativity checked in 4.5 makes these local isomorphisms agree, so no global quasi-projective factorization is required. □

Exercise 4.7

> **Problem.** For a finite flat projective morphism $f : X \to Y$, (a) construct the map $\lambda : \mathcal{O}_X \to \omega_f$ from $b \mapsto \mathrm{Tr}_{B/A}(b\cdot)$. (b) For a finite k-algebra E, compare etaleness, existence of an element of trace 1, and perfection of the trace pairing. (c),(d) Prove λ is an isomorphism exactly when f is etale. (e) If X, Y are integral and f generically separable, prove λ injective with cokernel supported on an empty set or a pure codimension-one closed subset.

Solution. (a) On $\operatorname{Spec} A \subset Y$ put $B = \mathcal{O}_X(f^{-1} \operatorname{Spec} A)$, finite locally free over A. Finite duality gives ω_f the module $B^\vee = \mathrm{Hom}_A(B, A)$ with action $(b\phi)(c) = \phi(bc)$. The trace pairing is B-linear for this action and commutes with localization, so the maps glue to λ and to its pushforward ρ.

(b) **The trace-1 condition is too weak.** For example over a characteristic-zero field $E = k[\epsilon]/(\epsilon^2)$ has $\mathrm{Tr}(1/2) = 1$ but is not etale. The correct equivalence is: E is etale if and only if its trace pairing is perfect. Nilpotents lie in the radical of that pairing since multiplication by a nilpotent has trace zero. Perfection forces E reduced, hence a product of fields. The pairing is perfect precisely when each field factor is separable, by the field trace theorem. Conversely this condition gives perfection. For nonzero etale E the trace functional is nonzero and therefore attains 1; the converse fails, as above. The zero algebra is a separate trivial case for the perfect-pairing equivalence.

(c) Formation of the pairing commutes with base change. On every fibre of an etale f it is perfect by (b). Nakayama makes ρ surjective on each local base ring, and a surjection of finite free modules of the same rank is an isomorphism. The corresponding map of B-modules, hence λ, is an isomorphism.

(d) Conversely an isomorphism remains one on every fibre. By (b) the fibres are etale. Together with finite flatness and finite presentation this says that f is etale.

(e) Over $K(Y)$ the pairing is perfect, so ρ is generically invertible. Both its source and target are locally free over integral Y; thus it is injective, and so is λ. To prove purity on X without assuming ω_f invertible, let $Q = B^\vee/B$ on an affine chart. As an A-module it has a presentation $0 \to A^n \to A^n \to Q \to 0$ with nonzero determinant. At a minimal prime $\mathfrak{p}$ of its support, the local determinant is a nonunit nonzerodivisor; the principal ideal theorem gives $\operatorname{ht} \mathfrak{p} = 1$.

For purity upstairs one needs to ensure that *each* minimal support prime of Q lies over such a height-one prime, rather than just knowing the image. This follows by localizing a finite algebra near its fibre factors after a faithfully flat etale base change: a selected factor is again finite locally free, its trace pairing has its own nonzero determinant, and its non-isomorphism locus on the base is a principal divisor. Each such factor's minimal support points lie over its divisor's generic points. Flat going-down and integrality then give height one upstairs. Equivalently this is the local discriminant argument for finite locally free algebras, applied factor by factor. Thus the closed support is empty or pure of codimension one. □

Exercise 4.8

Problem. For a finite injective map of Noetherian domains $A \to B$ with separable fraction-field extension L/K, put $W_{B/A} = \{\beta \in L : \mathrm{Tr}_{L/K}(\beta B) \subseteq A\}$. Prove $W_{B/A} \simeq \mathrm{Hom}_A(B, A)$ by the trace pairing. For Dedekind domains this is the codifferent.

Solution. Localizing gives $B \otimes_A K = L$: it is a finite-dimensional integral K-algebra, hence a field, and has fraction field L. An A-linear $\phi : B \to A$ extends uniquely to a K-linear map $L \to K$. Nondegeneracy of the separable field trace gives a unique $\beta \in L$ with $\phi(b) = \mathrm{Tr}_{L/K}(\beta b)$. That its original values lie in A is exactly the defining condition on $W_{B/A}$. Conversely that condition restricts the trace functional to $B \to A$. These constructions are inverse and B-linear. In the Dedekind case the resulting nonzero finite torsion-free rank-one B-module is a fractional ideal, giving the asserted sheaf description of finite duality. □

Exercise 4.9

Problem. (a) If $A \subseteq B \subseteq \mathrm{Frac}(A)$, with total fractions formed by nonzerodivisors, prove $\phi \mapsto \phi(1)$ identifies $\mathrm{Hom}_A(B, A)$ with the conductor $\{a \in A : aB \subseteq A\}$. (b) For a finite projective birational morphism of locally Noetherian integral schemes, deduce a canonical injection $\omega_f \to \mathcal{O}_X$.

Solution. (a) For $b = c/s \in B$ with s a nonzerodivisor of A, $s\phi(b) = \phi(c) = c\phi(1)$. In the total fraction ring this gives $\phi(b) = b\phi(1)$. Thus $a = \phi(1) \in A$ satisfies $aB \subseteq A$. Conversely multiplication by any such a gives the required map. The conductor is an ideal of both rings: $b(aB) \subseteq aB \subseteq A$ and $ab \in A$, for $b \in B$.

(b) On an affine open in Y, finite birationality identifies its inverse-image ring with a finite subring of the common function field. Finite duality and (a) identify ω_f with the conductor ideal in that ring. The inclusions are given by evaluation at 1 and commute with localization, so glue canonically. □

Exercise 4.10

Problem. If $f : X \to Y$ is projective with an r-dualizing sheaf, prove that a flat base change $Y' \to Y$ has dualizing sheaf $p^*\omega_f$.

Solution. We prove the locally Noetherian version, for coherent sheaves (and hence their quasi-coherent direct limits where the defining duality extends). The appropriate general result is flat base change for projective duality; here is its projective-space construction. Locally write $X \xrightarrow{i} \mathbf{P}^n_A \xrightarrow{g} \mathrm{Spec}\, A$ with i closed. Projective duality realizes the top dualizing sheaf for fibres of dimension at most r as

$$i_*\omega_f = \mathcal{E}xt^{n-r}_{\mathbf{P}^n_A}(i_*\mathcal{O}_X, \mathcal{O}_{\mathbf{P}^n_A}(-n-1)),$$

with the trace induced from the projective-space trace. This is the closed-immersion construction of the top dualizing sheaf, not an appeal to Theorem 6.4.32. A flat extension $A \to A'$ commutes

with this Ext sheaf: compute locally by a resolution by finite free modules (which may continue indefinitely); flat tensor product preserves its homology. It also pulls $\mathcal{O}(-n-1)$ to the same twist and carries the projective-space trace to its base change. Thus the construction on $X_{A'}$ is $p^*\omega_f$ with the base-changed trace. Uniqueness of dualizing sheaves identifies it with the desired one. The identifications agree on base overlaps. For unrestricted non-Noetherian Y' an extension of this duality construction is needed and is not established here. □

Exercise 4.11

> **Problem.** Let $X = V_+(F_1, \dots, F_{d-r}) \subset P = \mathbf{P}^d_A$ over Noetherian A, with every fibre pure of dimension r. Assume the homogeneous equations have positive degrees. Without Theorem 6.4.32: (a) prove flatness and fibrewise complete-intersection structure; (b) with $Z = V_+(F_2, \dots, F_{d-r})$ prove adjunction $\omega_{X/A} = \mathcal{O}_X(s-d-1)$, $s = \sum \deg F_i$, and the two restriction sequences; (c) prove local freeness of $R^p f_* \mathcal{O}_X(m)$ for $p = 0, r$, vanishing for $0 < p < r$, and $H^0(X, \mathcal{O}_X) = A$ for $r \geq 1$; (d) construct the trace and prove r-duality by induction.

Solution. (a) In each projective fibre the expected codimension is $c = d - r$. Its homogeneous equations form a regular sequence in the polynomial ring over the residue field: their ideal has height c and the ambient ring is Cohen–Macaulay. The fibrewise regular-sequence lifting criterion in the A-flat polynomial ring now gives a regular sequence and an A-flat quotient, locally along the resulting projective scheme. Thus X/A is flat and its fibres are complete intersections. This uses the corrected Cohen–Macaulay criterion in 3.4, not the false unrestricted exercise 3.5.

(b) Removing the first equation gives fibres of dimension $r + 1$. Indeed the remaining $c - 1$ equations have codimension at most $c-1$, while adding one equation can raise codimension by at most one. No extra projective component can avoid the positive-degree hypersurface: a positive-dimensional projective component meets every positive-degree hypersurface. Consequently those fibres are complete intersections too. Multiplication by F_1 is injective on $\mathcal{O}_Z(-d_1)$, where $d_1 = \deg F_1$, and adjunction for its regular immersion gives

$$\omega_{X/A} \simeq \omega_{Z/A}|_X \otimes \mathcal{O}_X(d_1).$$

Starting with $\omega_{P/A} = \mathcal{O}_P(-d-1)$ proves $\omega_{X/A} = \mathcal{O}_X(s-d-1)$; the printed $s-n-1$ has $n = d$. The exact sequences are

$$0 \to \mathcal{O}_Z(m - d_1) \to \mathcal{O}_Z(m) \to \mathcal{O}_X(m) \to 0,$$

$$0 \to \omega_{Z/A}(-m) \to \omega_{Z/A}(-m + d_1) \to \omega_{X/A}(-m) \to 0.$$

The second is the first restriction sequence for $\omega_Z(-m + d_1)$ combined with adjunction.

(c) Use the graded Koszul resolution on the F_i and the usual cohomology of $\mathcal{O}_{\mathbf{P}^d}(j)$. On every field fibre this gives no cohomology in degrees $0 < p < r$, and $H^0(\mathcal{O}) = k$ for $r \geq 1$. For $r > 0$, its degree-m global sections have dimension given by the degree-m coefficient of

$$\frac{\prod_i (1 - t^{d_i})}{(1-t)^{d+1}}.$$

The Euler characteristic is likewise determined by the Koszul resolution and the degrees, so the dimension of $H^r(\mathcal{O}(m))$ is constant too. For $r = 0$, the finite flat scheme has degree $\prod_i d_i$, and

all its invertible twists have that constant H^0 dimension. Cohomology and base change for the flat projective family now makes the asserted endpoint direct images locally free and gives the intermediate vanishings. The unit map $A \to H^0(X, \mathcal{O}_X)$ is an isomorphism for $r \geq 1$, since it is an isomorphism on every fibre between locally free modules of rank one. This argument works over a nonreduced base as well.

(d) Induct on c, starting with duality on projective space. Suppose Z already has its $(r+1)$-dualizing sheaf and trace. The connecting map of the second sequence at $m = 0$ gives $R^r f_* \omega_X \to R^{r+1} g_* \omega_Z$; compose with tr_g to define tr_f.

Here is a precise way to check the diagram and the dimension-zero endpoint at once. For the regular divisor $i : X \to Z$ the two-term resolution of $i_* \mathcal{O}_X$ gives

$$\mathcal{H}om_Z(i_* \mathcal{O}_X, \omega_Z) = 0, \qquad \mathcal{E}xt^1_Z(i_* \mathcal{O}_X, \omega_Z) = i_* \omega_X,$$

with no other Ext sheaves. The first equality uses injectivity of multiplication by F_1 on the invertible sheaf ω_Z. For any coherent $\mathcal{F}$ on X, the change-of-rings spectral sequence therefore identifies

$$\mathrm{Hom}_X(\mathcal{F}, \omega_X) = \mathrm{Ext}^1_Z(i_* \mathcal{F}, \omega_Z).$$

The derived projective-space duality calculation, applied to the Koszul resolution of Z, identifies this last group with $\mathrm{Hom}_A(H^r(X, \mathcal{F}), A)$ in the top-degree situation: its other possible contribution is $\mathrm{Ext}^1_A(H^{r+1}(X, \mathcal{F}), A) = 0$, since the latter cohomology vanishes. This may also be obtained directly by the top-degree duality spectral sequence. The trace is exactly the connecting-map trace above, as seen from the two-term resolution. Hence it proves the required duality for all coherent $\mathcal{F}$, including $r = 0$, without using Theorem 6.4.32.

For the diagram requested in the print, its top row is the beginning of the first long exact sequence, and its bottom row is the dual of the endpoint segment of the second. Multiplication by F_1 is adjoint to multiplication by F_1, and the connecting maps are adjoint by the trace definition. Thus it commutes, with compatible choices of connecting-map signs. The two left arrows are injective; for the bottom arrow this follows by dualizing the surjection in top cohomology. When $r \geq 1$, intermediate cohomology vanishes, making the right arrows surjective. When $r = 0$, the continuation of these sequences supplies the additional terms; the Ext calculation above proves precisely the endpoint identification which cannot be obtained by omitting those terms. This proof uses derived duality on projective space and its elementary regular-immersion change-of-rings calculation. Thus it avoids Theorem 6.4.32, but does use stronger homological tools than the elementary diagram requested in the exercise. □